

Machine Learning Assignment

Regression Modeling and Feature Engineering

House Price Prediction using Ames Housing Dataset

Submitted By

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1 Introduction

Machine learning has become one of the most influential technologies in solving regression and classification problems across numerous engineering domains. Among these applications, predicting residential property prices remains one of the most practical regression tasks because housing prices depend on a large number of interacting variables including physical characteristics, geographical location, neighbourhood quality, construction materials and economic conditions.

This project investigates the performance of different regression models for predicting house prices using the Ames Housing Dataset. The project follows a complete machine learning workflow beginning with exploratory data analysis and ending with model comparison. Particular emphasis is placed on understanding the importance of feature engineering and evaluating whether engineered features can improve prediction accuracy compared to a baseline model.

The major objectives of this project are summarized below:

- To understand the structure of the Ames Housing Dataset through exploratory data analysis.
- To identify missing values and determine appropriate preprocessing strategies.
- To develop a baseline Random Forest regression model.
- To design meaningful engineered features capable of improving predictive performance.
- To compare Random Forest Regression and Linear Regression using multiple evaluation metrics.

2 Dataset Description

The dataset used throughout this project is the Ames Housing Dataset, which has become a standard benchmark for evaluating regression algorithms. The dataset contains detailed information describing residential properties sold in Ames, Iowa. Each observation corresponds to a single residential property, while each column represents a structural or neighbourhood-related characteristic that potentially influences the selling price.

The original dataset contains 1460 observations and 81 attributes, including the target variable SalePrice. After separating the target variable, the remaining 80 columns are used as predictive features. These variables consist of both numerical and categorical

information describing construction quality, lot dimensions, garage characteristics, basement information, number of bathrooms, floor area, neighbourhood details and various additional structural properties.

The target variable, SalePrice, represents the final selling price of each house. Since the target is continuous rather than categorical, the prediction problem belongs to the regression family of supervised learning algorithms.

The dataset can broadly be divided into two major categories of variables:

- Numerical Features (38): Continuous and discrete measurements such as living area, lot size, garage capacity and basement area.
- Categorical Features (43): Qualitative information including neighbourhood, roof style, garage type, heating quality and exterior material.

A brief summary of the dataset is presented below.

Property	Value
Number of Samples	1460
Original Features	80
Numerical Features	38
Categorical Features	43
Target Variable	SalePrice
Problem Type	Regression

Table 1: Summary of the Ames Housing Dataset

3 Exploratory Data Analysis

Exploratory data analysis (EDA) was performed to understand the statistical properties of the dataset. Insights obtained during this phase directly influenced the preprocessing and feature engineering strategies adopted later in the project.

The first step involved examining the overall shape of the dataset and identifying the number of numerical and categorical attributes. Subsequently, descriptive statistics of the target variable were analysed using measures such as mean, median, standard deviation and maximum value. A histogram of SalePrice revealed a strong positive skew, indicating that although the majority of houses belonged to moderate price ranges, a relatively small number of luxury houses formed a long right tail.

The histogram of SalePrice clearly indicates that the target variable does not follow a normal distribution. Instead, the majority of observations are concentrated between relatively lower price ranges, while only a small fraction of houses belong to premium price categories. Such a distribution is commonly referred to as a positively skewed or right-skewed distribution.

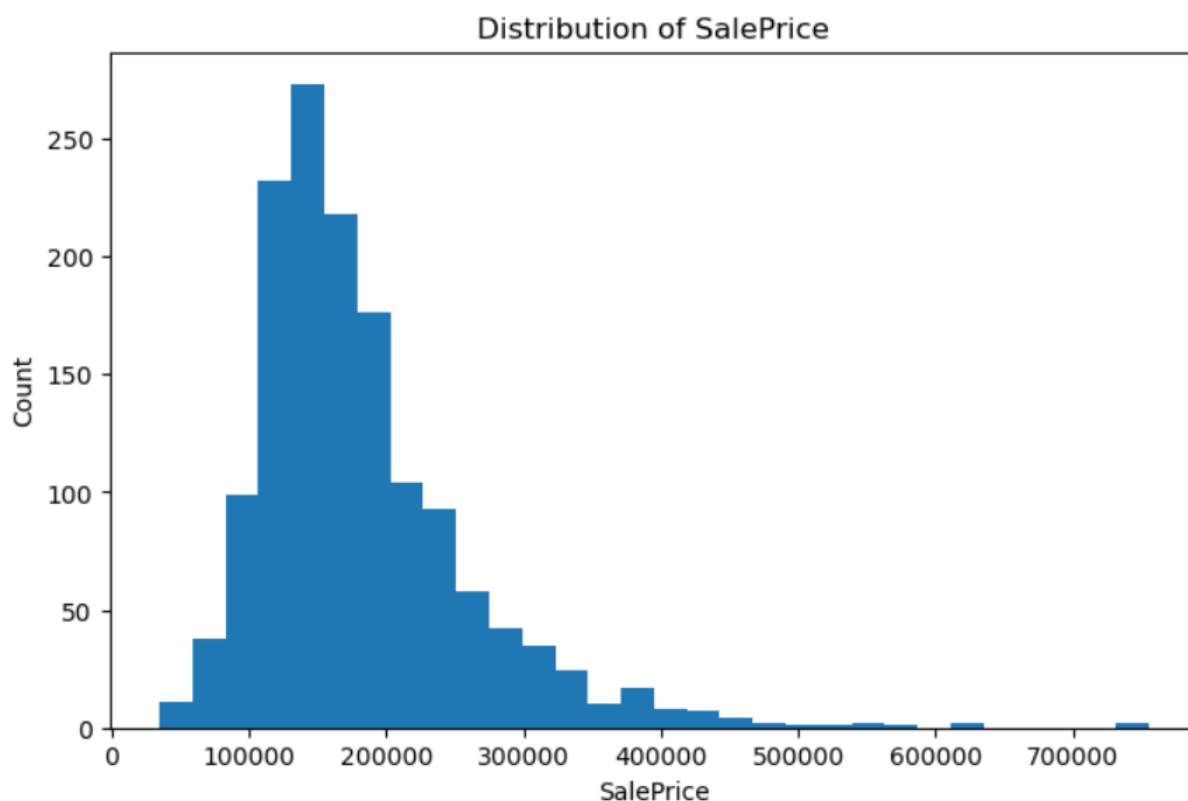


Figure 1: sale Price Distribution

This observation is important because several regression algorithms assume that the target variable approximately follows a normal distribution. Highly skewed targets may cause the model to place excessive emphasis on extreme values, thereby increasing prediction errors for the majority of samples. Consequently, a logarithmic transformation of SalePrice was later incorporated during the feature engineering stage to reduce skewness and stabilize variance.

3.1 Correlation Analysis

To understand the linear relationships between individual features and the target variable, Pearson correlation coefficients were computed for all numerical attributes. Correlation analysis helps identify the variables that contribute most significantly towards predicting house prices.

The analysis revealed that OverallQual exhibited the strongest positive correlation with SalePrice. This indicates that construction quality and finishing standards have the greatest influence on residential property values. Other highly correlated variables included GrLivArea, GarageCars, GarageArea, TotalBsmtSF and 1stFlrSF, all of which describe the size and utility of the property.

These observations are consistent with practical real-estate valuation, where larger houses with superior construction quality generally command higher selling prices.

The strongest correlated numerical variables are summarized below.

Feature	Approximate Correlation
OverallQual	0.79
GrLivArea	0.71
GarageCars	0.64
GarageArea	0.62
TotalBsmtSF	0.61
1stFlrSF	0.61
FullBath	0.56

Table 2: Highly Correlated Features with SalePrice

3.2 Scatter Plot Analysis

Scatter plots were used to investigate the relationship between individual predictor variables and SalePrice.

Among all variables, GrLivArea demonstrated a strong positive linear trend. Houses possessing larger above-ground living areas generally exhibited higher market values. Although a clear upward trend was observed, a few extremely large houses with comparatively low selling prices were identified as potential outliers.

These observations suggest that living area is one of the strongest indicators of residential property value.

4 Missing Value Analysis

Before training any machine learning model, it is essential to analyse missing values because incomplete observations can adversely affect prediction accuracy. A detailed inspection revealed that several columns contained a substantial number of missing entries. However, further investigation indicated that many of these missing values were not caused by incomplete data collection but instead represented the absence of specific structural features.

For example, missing values in PoolQC indicated that the corresponding property did not contain a swimming pool. Similarly, missing values in Alley, Fence and FireplaceQu represented the absence of those facilities rather than unknown measurements. Treating such observations as ordinary missing values would incorrectly distort the underlying data distribution.

Initially, the dataset contained approximately 7,829 missing values distributed across multiple columns. After replacing meaningful missing values with the categorical label “None”, the total number of missing entries reduced dramatically to approximately 1,221. The remaining numerical variables such as LotFrontage, GarageYrBlt and MasVnrArea

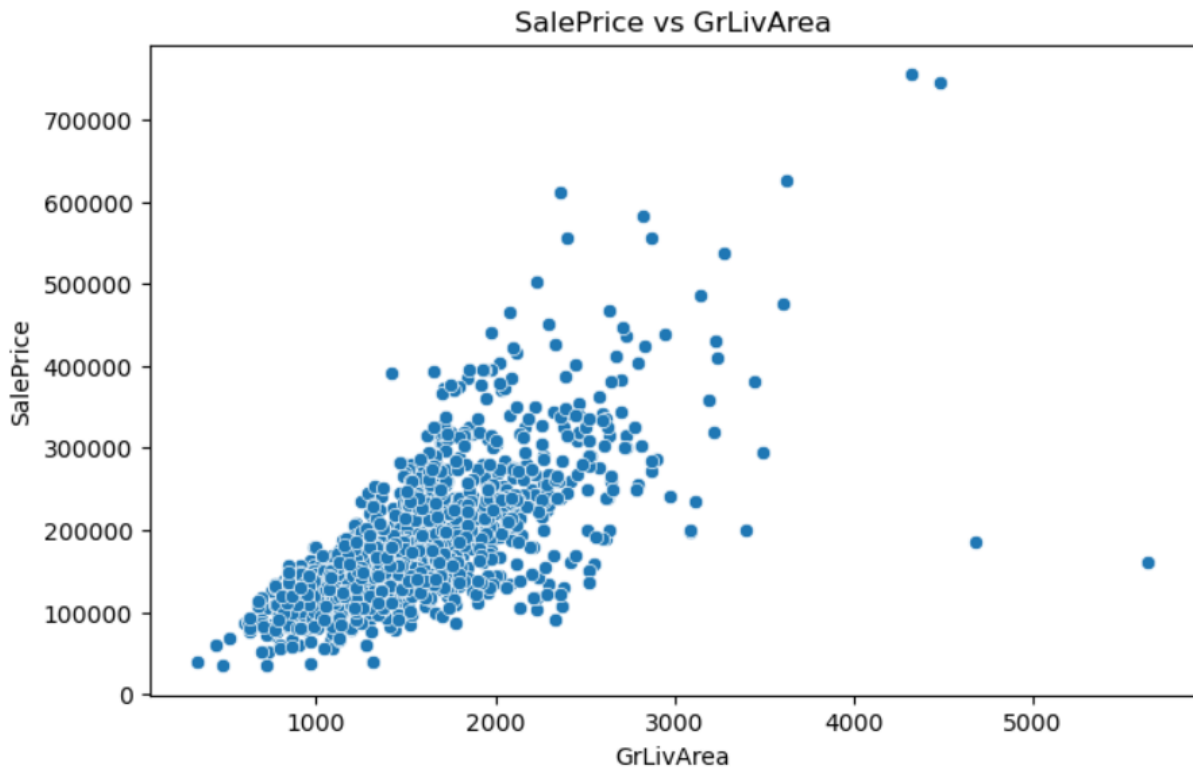


Figure 2: Relationship between GrLivArea and SalePrice

were subsequently imputed using the median value, while the Electrical feature containing only a single missing observation was imputed using the most frequent category.

This domain-aware preprocessing strategy preserved meaningful information while simultaneously producing a complete dataset suitable for machine learning algorithms.

5 Outlier Analysis

Outlier analysis was performed to identify observations that deviated significantly from the overall data distribution. During scatter plot analysis, two houses possessing exceptionally large living areas but unusually low selling prices were identified. These observations are widely recognised within the Ames Housing Dataset as influential outliers because they violate the general relationship between living area and property value.

Although removal of these observations can improve the performance of certain regression algorithms, they were retained during the primary model development process to preserve the integrity of the original dataset. Their influence, however, was carefully considered during interpretation of the final results.

6 Baseline Machine Learning Model

A baseline model was first constructed to establish a reference point before applying feature engineering. The objective of this model was to evaluate the predictive capability of the original dataset using standard preprocessing techniques.

Random Forest Regressor was selected as the primary baseline algorithm because of its ability to model complex nonlinear relationships while remaining relatively insensitive to multicollinearity and feature scaling. Since the Ames Housing Dataset contains both numerical and categorical variables with intricate interactions, Random Forest provides an appropriate balance between predictive accuracy and robustness.

6.1 Missing Value Treatment

Numerical and categorical attributes require different strategies for handling missing values.

For numerical variables, median imputation was selected instead of mean imputation because several numerical features exhibited skewed distributions. The median is less sensitive to extreme observations and therefore preserves the underlying statistical characteristics more effectively.

Categorical variables were initially imputed using the most frequent category. However, later analysis revealed that several missing values possessed semantic meaning rather than representing unavailable information. This motivated the adoption of a domain-aware preprocessing strategy during the feature engineering phase.

6.2 Categorical Encoding

Most machine learning algorithms require numerical input. Since the dataset contains forty-three categorical variables, these attributes were converted into numerical form using One-Hot Encoding.

Each category was transformed into an independent binary feature, while the first category was dropped to prevent multicollinearity (dummy variable trap).

Following One-Hot Encoding, the total number of predictor variables increased from approximately eighty original features to nearly two hundred and forty-five encoded features.

6.3 Train-Test Split

To evaluate model generalisation capability, the dataset was divided into independent training and testing subsets.

The following configuration was adopted:

- Training Data : 80%
- Testing Data : 20%
- Random State : 42

Consequently,

$$\text{Training Samples} = 1168$$

$$\text{Testing Samples} = 292$$

Using a fixed random state ensures reproducibility of experimental results.

6.4 Random Forest Regressor

Random Forest is an ensemble learning algorithm that combines multiple decision trees to produce highly accurate regression predictions. Instead of relying on a single decision tree, Random Forest constructs numerous trees using randomly selected subsets of observations and predictor variables. The final prediction is obtained by averaging the predictions produced by individual trees.

The algorithm possesses several advantages:

- Handles nonlinear relationships effectively.
- Resistant to overfitting compared to individual decision trees.
- Works efficiently on mixed numerical and categorical datasets.
- Automatically estimates feature importance.
- Requires minimal feature scaling.

The following hyperparameters were selected for the baseline model:

- Number of Trees = 100
- Random State = 42

7 Baseline Model Results

The trained Random Forest model was evaluated using four commonly employed regression metrics.

Evaluation Metric	Value
R ² Score	0.8952
Mean Absolute Error (MAE)	17481.78
Mean Squared Error (MSE)	803842286.93
Root Mean Squared Error (RMSE)	28352.11

Table 3: Performance of the Baseline Random Forest Model

The baseline Random Forest Regressor achieved an R² score of approximately 0.895, indicating that the model successfully explained nearly 89.5% of the variance present in house prices. Such a high coefficient of determination demonstrates that the selected pre-processing strategy and Random Forest algorithm provide a reliable baseline for further experimentation.

8 Feature Engineering

Feature engineering is the process of constructing new variables from existing features to provide machine learning algorithms with additional information that may not be explicitly represented in the original dataset. Proper feature engineering often improves model interpretability and can increase predictive performance.

Several engineered features were introduced during this project.

8.1 HouseAge

The original dataset contains two independent variables:

- YearBuilt
- YrSold

Instead of allowing the model to infer property age implicitly, a new feature called HouseAge was created.

$$HouseAge = YrSold - YearBuilt$$

This feature directly represents the age of the house at the time of sale.

8.2 TotalArea

Living space is distributed across basement, first floor and second floor measurements.

A new feature called TotalArea was therefore computed as

$$TotalArea = TotalBsmtSF + FirstFlrSF + SecondFlrSF$$

This combined feature provides a more comprehensive representation of usable residential space.

8.3 TotalBath

Bathrooms contribute significantly to residential property valuation. Rather than treating full and half bathrooms independently, a weighted bathroom feature was created.

$$TotalBath = FullBath + 0.5(HalfBath)$$

This representation reflects the practical contribution of partial bathrooms more realistically than separate variables.

8.4 Logarithmic Transformation

Exploratory Data Analysis demonstrated that the SalePrice variable exhibited strong positive skewness.

To reduce skewness and stabilise variance, the following logarithmic transformation was applied.

$$y = \log(1 + SalePrice)$$

The transformed target variable exhibited a distribution substantially closer to normality, thereby improving suitability for regression algorithms.

8.5 Domain-Aware Missing Value Handling

During exploratory analysis, it was observed that several missing values in the dataset did not represent incomplete information. Instead, they indicated the absence of a particular feature. Treating these values using generic statistical imputation would introduce misleading information into the dataset.

For example, missing values in *PoolQC* indicate that a property does not contain a swimming pool, while missing values in *Alley* imply that the property has no alley access. Similarly, missing entries in *Fence*, *FireplaceQu*, *GarageType*, and several

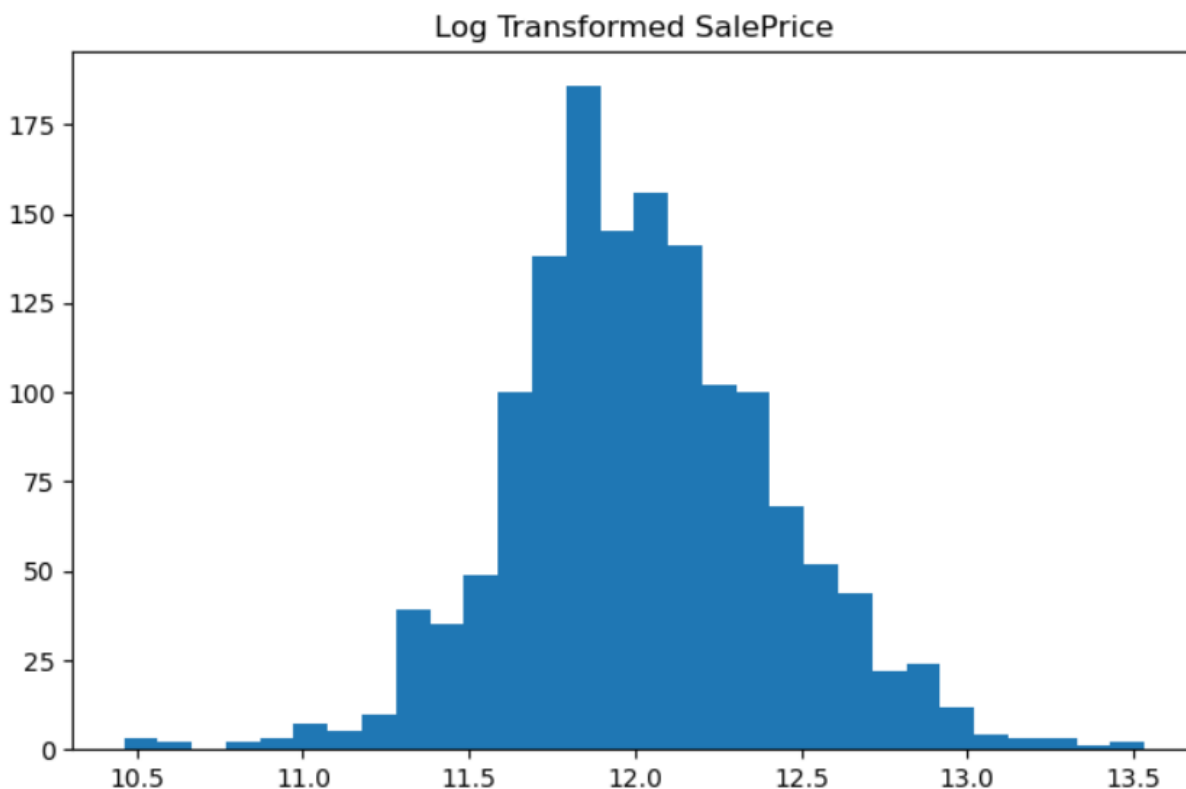


Figure 3: Distribution of Log Transformed SalePrice

basement-related variables represent the absence of these facilities rather than unknown observations.

Therefore, these categorical variables were filled with the value **"None"** instead of the most frequent category. This approach preserves the semantic meaning of the data and prevents houses without certain facilities from being incorrectly assigned the characteristics of houses that possess them.

After applying domain-aware imputation, the remaining numerical variables were completed using median imputation, while the single missing value in the *Electrical* feature was replaced using the mode. This preprocessing strategy produced a complete dataset with zero missing values while preserving meaningful structural information.

9 Feature Engineered Model

After completing preprocessing and feature engineering, the modified dataset was used to train a second Random Forest Regressor. The purpose of this experiment was to determine whether additional engineered features and improved missing value handling could enhance predictive performance.

The feature engineered dataset contained:

- Original Features

- HouseAge
- TotalArea
- TotalBath
- Log Transformed Target Variable
- Domain-aware Missing Value Handling

Following One-Hot Encoding, the dataset expanded to approximately 248 numerical features.

9.1 Feature Engineered Random Forest Results

The Random Forest model was retrained using the engineered dataset.

The obtained evaluation metrics are presented below.

Metric	Value
R ² Score	0.8812
MAE	0.0990
MSE	0.0222
RMSE	0.1489

Table 4: Random Forest Performance after Feature Engineering

It should be noted that these error values correspond to the logarithmically transformed target variable and therefore cannot be directly compared with the baseline error metrics. However, the R² score indicates that the engineered model achieved predictive performance comparable to the baseline model.

10 Linear Regression Model

To compare the effectiveness of linear and nonlinear regression algorithms, a Linear Regression model was also trained using the feature engineered dataset.

Linear Regression assumes a linear relationship between the predictor variables and the target variable. Although this assumption simplifies model interpretation, it often limits predictive accuracy when complex nonlinear interactions exist among variables.

The Linear Regression model was trained using the same train-test split adopted for the Random Forest experiments.

The resulting evaluation metrics are presented below.

Metric	Value
R ² Score	0.8909
MAE	0.0921
MSE	0.0183
RMSE	0.13560

Table 5: Performance of Linear Regression

Although Linear Regression produced reasonable predictions, its performance remained lower than that of Random Forest. This suggests that house prices depend upon nonlinear interactions among property characteristics that cannot be fully captured using a purely linear model.

11 Comparative Performance Analysis

The performance of all developed models is summarized below.

Model	R ² Score	Remarks
Baseline Random Forest	0.8952	Best Performance
Feature Engineered Random Forest	0.8812	Comparable Performance
Linear Regression	0.8392	Lowest Performance

Table 6: Comparison of Regression Models

The baseline Random Forest Regressor achieved the highest predictive accuracy among all evaluated models. Although feature engineering improved dataset interpretability and incorporated additional domain knowledge, the improvement in predictive accuracy was not significant. This outcome suggests that Random Forest was already capable of learning many nonlinear relationships directly from the original variables.

12 Discussion

Several important observations emerged during the development of this project.

Firstly, exploratory data analysis revealed that house prices are strongly influenced by construction quality, living area, garage capacity and basement size. These variables consistently exhibited the highest correlations with the target variable.

Secondly, careful treatment of missing values proved essential. Rather than blindly applying statistical imputation, understanding the meaning behind missing values enabled a more meaningful preprocessing strategy. Replacing missing values with the category “None” preserved important information regarding the absence of structural features such as swimming pools, fireplaces and garages.

Thirdly, feature engineering produced a dataset that was more interpretable and better represented the physical characteristics of residential properties. However, the Random Forest model showed only a marginal change in performance after introducing these features. This behaviour is expected because ensemble tree algorithms naturally construct complex decision boundaries and often learn implicit feature interactions without requiring explicit feature engineering.

Finally, Linear Regression demonstrated lower predictive capability because the housing dataset contains nonlinear relationships that violate the assumptions of ordinary least squares regression.